

EE227 – Digital Logic Design

Lecture 26

Outline

- Ripple Counters
 - Binary Ripple Counter
 - BCD Ripple Counter
- Synchronous Counter
 - Binary Counter
 - Binary Up/Down Counter

Counters

- A register that goes through a prescribed sequence of states upon the application of input pulses is called a counter
- Counters are available in two categories
- Ripple counters
 - In a ripple counter, a flip-flop output transition serves as a source for triggering other flip-flops
 - Clock is not common
- Synchronous counters
 - In a synchronous counter, the C inputs of all flip-flops receive the common clock

Ripple Counters

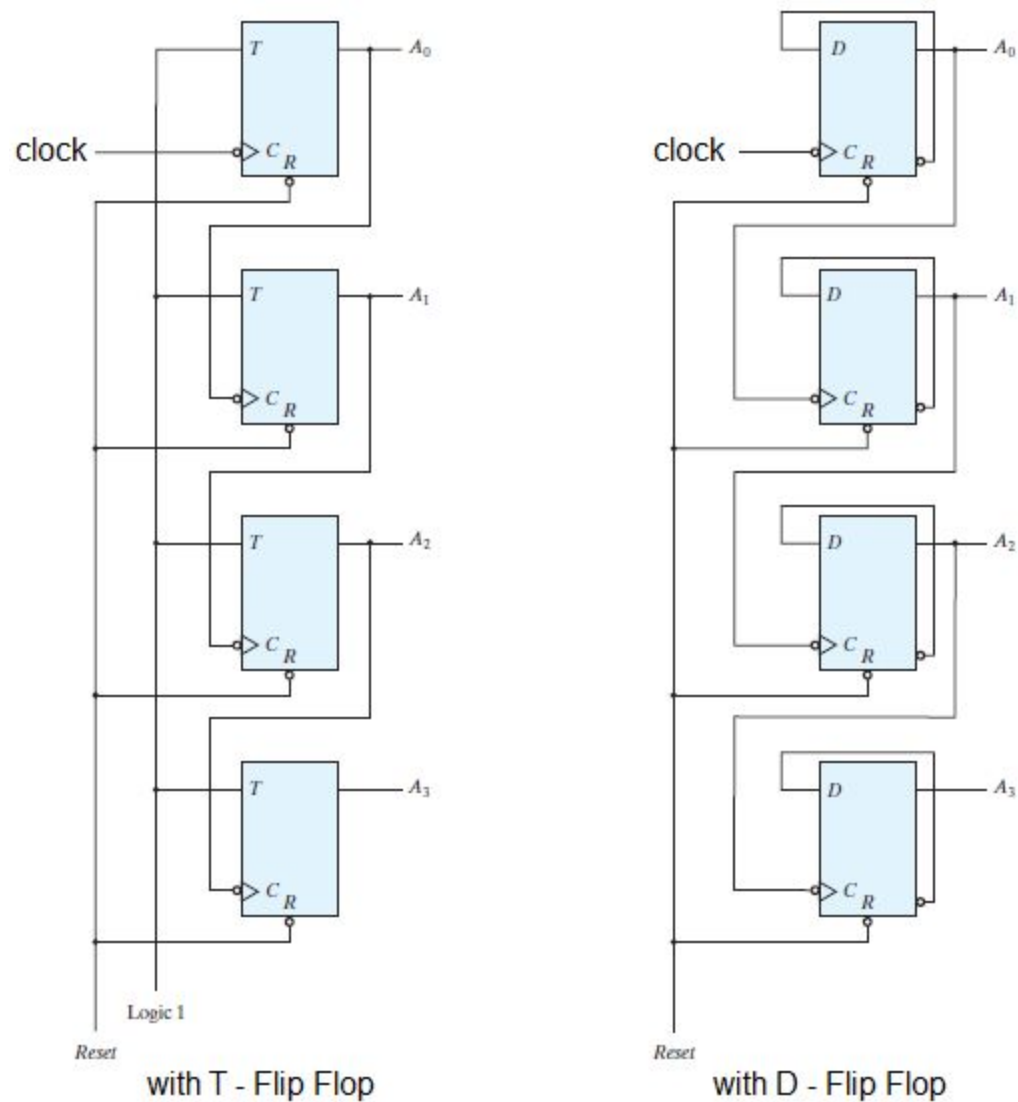
- In a ripple counter, a flip-flop output transition serves as a source for triggering other flip-flops
- Clock is not common
- We will discuss two types of ripple counters
- Binary Ripple Counter
 - A counter that follows the binary number sequence (0, 1, 2, 3, ...)
- BCD Ripple Counter
 - A BCD counter follows a sequence of 10 states and returns to 0 after the count of 9

4 – Bit Binary Ripple Counter (1/2)

- 4 bits $\Rightarrow$ 4 flip flops are required
- LSB (A_0) is complementing every time
 - Connect the flip flop producing A_0 directly to a clock
- A_1 is changing when A_0 changes from 1 to 0 (at negative edge)
 - A_0 is connected to the clock of flip flop producing A_1
- A_2 is changing when A_1 changes from 1 to 0 (at negative edge)
 - A_1 is connected to the clock of flip flop producing A_2
- A_3 is changing when A_2 changes from 1 to 0 (at negative edge)
 - A_2 is connected to the clock of flip flop producing A_3

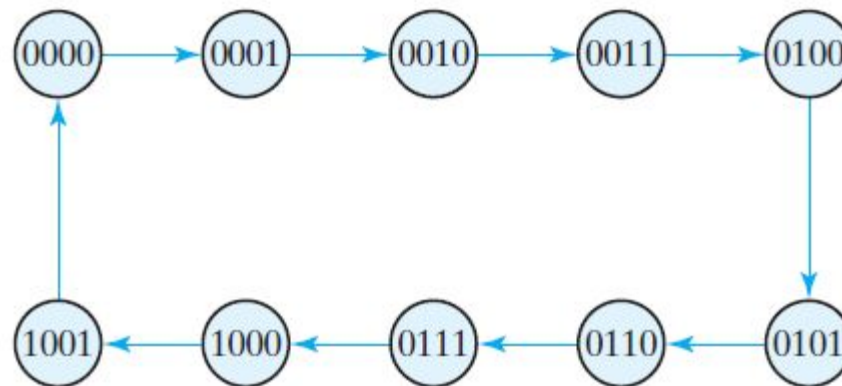
In decimal	4 – bit binary number			
	A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

4 – Bit Binary Ripple Counter (2/2)



BCD Ripple Counter (1/3)

- A BCD counter follows a sequence of 10 states and returns to 0 after the count of 9
- 10 states $\Rightarrow$ at least 4 flip flops are required
- State diagram is shown below



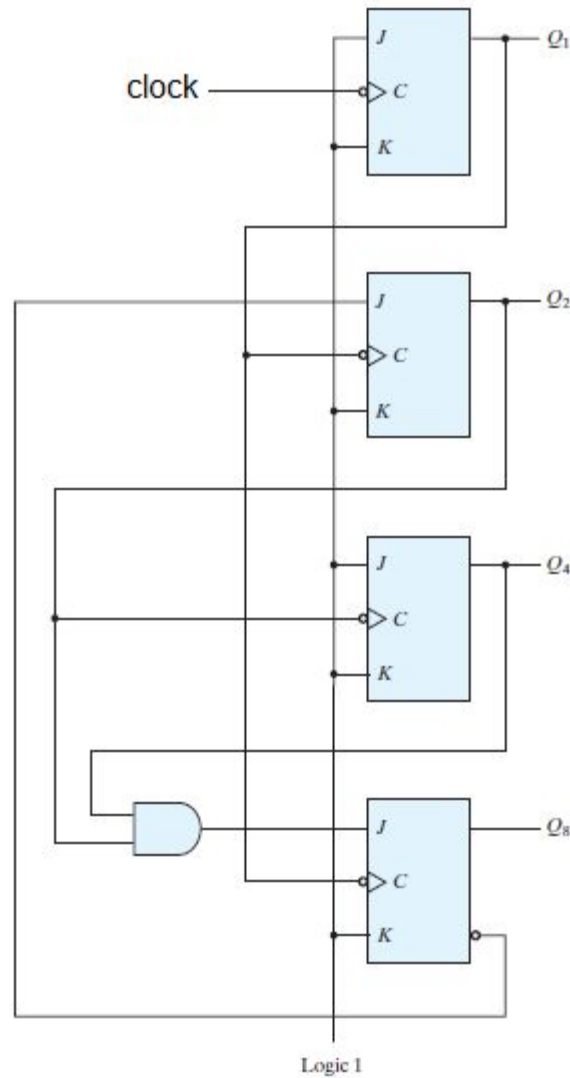
BCD Ripple Counter (2/3)

- Q_1 is complemented on every clock pulse
- Q_2 is complemented if $Q_8 = 0$ and Q_1 goes from 1 to 0
- Q_2 is clear if $Q_8 = 1$ and Q_1 goes from 1 to 0
- Q_4 is complemented if Q_2 goes from 1 to 0
- Q_8 is complemented Q_1 goes from 1 to 0
- Q_8 is cleared if either Q_4 or Q_2 is 0 and Q_1 goes from 1 to 0

Decimal	BCD Code			
	Q_8	Q_4	Q_2	Q_1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

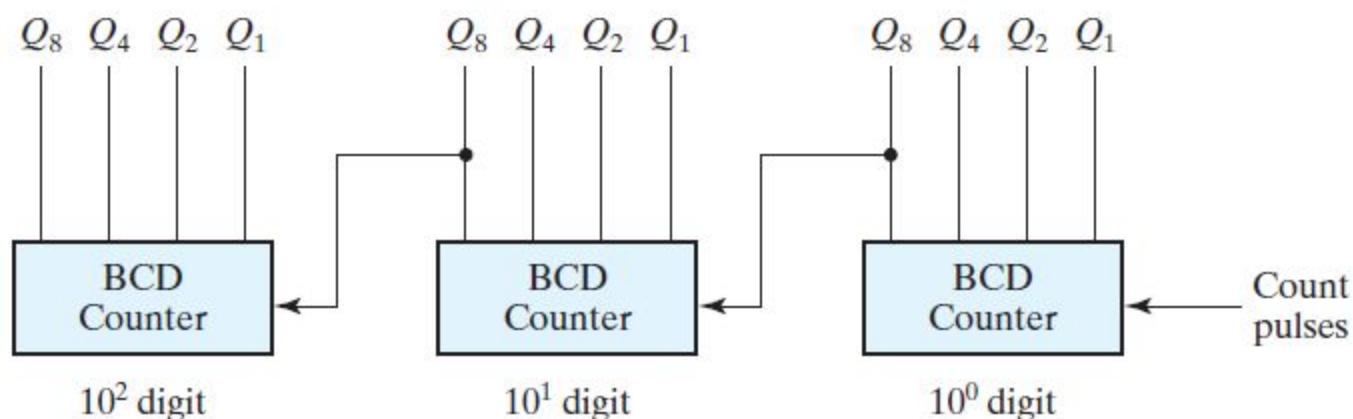
J	K	$Q(t+1)$	
0	0	$Q(t)$	No Change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

BCD Ripple Counter (3/3)



BCD Decade Counter

- The BCD counter discussed previously is a decade counter as it counts from 0 to 9
- To count in decimal from 0 to 99, we need a two-decade counter
- To count from 0 to 999, we need a three-decade counter
- Multiple decade counters can be constructed by connecting BCD counters in cascade, one for each decade



Synchronous Counters

- Synchronous counters are different from ripple counters in that clock pulses are applied to the inputs of all flip-flops
- A common clock triggers all flip-flops simultaneously, rather than one at a time in succession as in a ripple counter
- The design procedure for synchronous counters was presented in chapter 5, and the design of a three-bit binary counter was carried out
- Here we will enhance the design for 4 bit binary counter

4 – Bit Synchronous Binary Up Counter

DC \ BA	00	01	11	10
00				
01			1	
11			1	
10				

$$T_D = CBA$$

DC \ BA	00	01	11	10
00			1	
01			1	
11			1	
10			1	

$$T_C = BA$$

DC \ BA	00	01	11	10
00		1	1	
01		1	1	
11		1	1	
10		1	1	

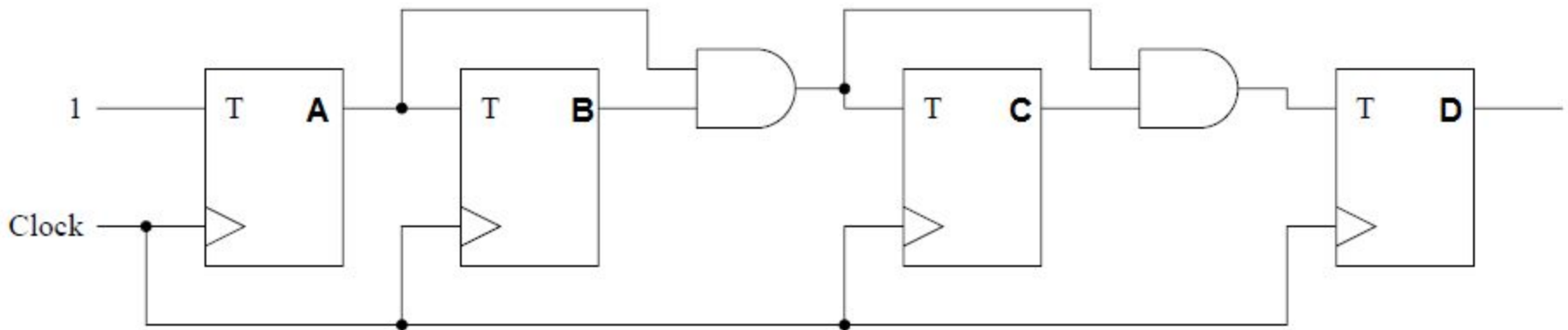
$$T_B = A$$

DC \ BA	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$T_A = 1$$

Present State				Next State				Flip Flop Inputs			
D	C	B	A	D	C	B	A	T_D	T_C	T_B	T_A
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	1	0	1	0	0	0	1	1
1	0	1	0	1	0	1	1	0	0	0	1
1	0	1	1	1	1	0	0	0	1	1	1
1	1	0	0	1	1	1	0	0	0	0	1
1	1	0	1	1	1	1	1	0	0	1	1
1	1	1	0	1	1	1	1	0	0	0	1
1	1	1	1	0	0	0	0	1	1	1	1

4 – Bit Synchronous Binary Up Counter



4 – Bit Synchronous Binary Down Counter

- From State Table, we get

$$T_D = A'B'C'$$

$$T_C = A'B'$$

$$T_B = A'$$

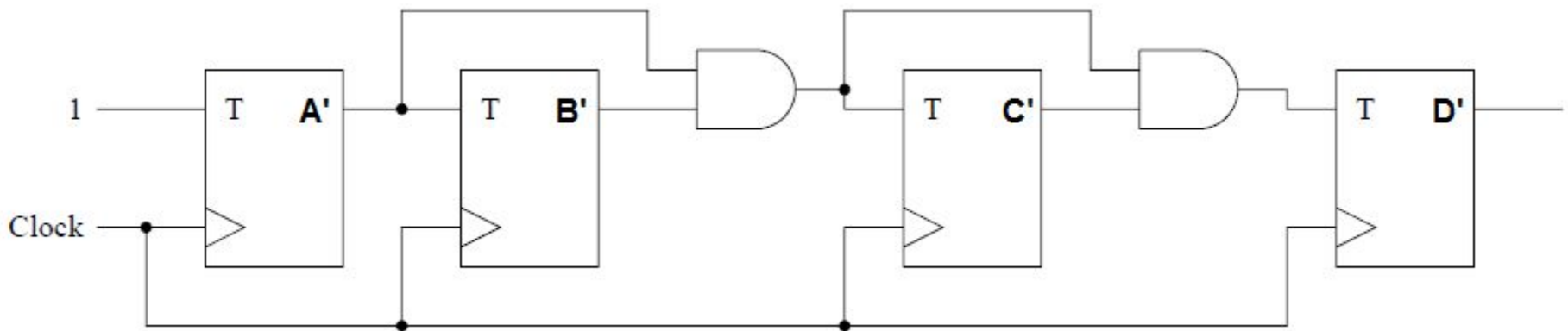
$$T_A = 1$$

Present State			
D	C	B	A
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Next State			
D	C	B	A
1	1	1	1
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Flip Flop Inputs			
T _D	T _C	T _B	T _A
1	1	1	1
0	0	0	1
0	0	1	1
0	0	0	1
0	1	1	1
0	0	0	1
0	0	1	1
0	0	0	1
1	1	1	1
0	0	0	1
0	0	1	1
0	0	0	1
0	1	1	1
0	0	0	1
0	0	1	1
0	0	0	1

4 – Bit Synchronous Binary Down Counter



3 Bit Binary Up/Down Counter (1/2)

By using K – Map

We Get

$$T_A = 1$$

$$T_B = M'Q_A + MQ'_A$$

$$T_C = M'Q_BQ_A + MQ'_BQ'_A$$

Control	Present State			Next State			Flip Flop Inputs		
M	Q _C	Q _B	Q _A	Q _C	Q _B	Q _A	T _C	T _B	T _A
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

4 Bit Binary Up/Down Counter

